More Backtracking Examples

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Subset Sum

TSP is an example of an ***NP-hard*** problem. Another is SUBSET-SUM:

Given a set of positive integers ***S*** and a target ***T***, is there a subset of ***S*** that adds up to ***T***?

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Given a set of positive integers ***S*** and a target ***T***, is there a subset of ***S*** that adds up to ***T***?

S = {6, 12, 8, 18, 3, 19}, ***T*** = 39

Subset Sum

Target: 39

Subset Sum

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S = {6, 12, 8, 18, 3, 19}, ***T*** = 39

YES, 8+12+19 = 39 or 3+6+12+18 = 39

Bin-packing, knapsack, partition, etc.

Subset Sum is one variant of a general class of “packing” problems.

- Bin-packing: fewest number of trips
- Knapsack: one trip, maximize value
- Subset sum: exactly at capacity
- and so on…

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Subset Sum is one variant of a general class of “packing” problems.

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- Knapsack: one trip, maximize value
- Subset sum: exactly at capacity
- and so on…

Most interesting variations are NP-hard!

Subset Sum

Given a set of positive integers ***S*** and a target ***T***, is there a subset of ***S*** that adds up to ***T***?

S = {6, 12, 8, 18, 3, 19}, ***T*** = 50

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NO SOLUTION

Subset Sum

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Goal: design an algorithm for solving Subset Sum

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Given a set of positive integers ***S*** and a target ***T***, is there a subset of ***S*** that adds up to ***T***?

Goal: design an algorithm for solving Subset Sum

Brute-force using backtracking: effectively try every subset

The set of subsets

If $\boldsymbol{S}$ is a set of $\boldsymbol{n}$ elements, how many subsets are there? (including $\boldsymbol{S}$ and $\varnothing$)

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Ans: 2^n

The set of subsets

If S is a set of n elements, how many subsets are there? (including S and $\varnothing$)

Ans: 2^n

Proof: we can pick a subset $X \subseteq S$ in the following way:

- For each element, either:
  - Include it
  - Exclude it

Two possible choices for each element, so there are 2^n elements.

Base cases

Two parameters:

- ***S***, a set of positive numbers
- ***T***, a target value

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Base cases: ***S*** is empty, and/or ***T*** ≤ 0.

Base cases

When ***S*** is empty, what values of ***T*** can it add up to?

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Ans: only subset is itself, which has sum 0.

$$
\begin{aligned}
&\textsc{Sum}(A[1,\ldots,n]):\\
&\qquad sum := \boxed{0}\\
&\qquad \textbf{for } i = 1,\ldots,n\\
&\qquad\qquad sum := sum + A[i]\\
&\qquad \textbf{return } sum
\end{aligned}
$$

Base cases

When $T \leq 0$, which sets S (of positive integers) have a subset that adds up to T?

Base cases

When $T \leq 0$, which sets S (of positive integers) have a subset that adds up to T?

- If $T < 0$, then nothing
- If $T = 0$, then every set S has the empty set as a subset.

Base cases

Coding up the base cases...

```
SubsetSum(S, T):
    if T = 0
        return True
    else if T < 0 or S = ∅
        return False
    else
        // some stuff here
```

Enumerating subsets

Let ***e*** be any element of ***S***. Every subset ***X*** of ***S*** either:

- has ***e***
- doesn't have ***e***

Enumerating subsets

Let ***e*** be any element of ***S***. Every subset ***X*** of ***S*** that sums to ***T*** either:

- has ***e*** — ***X*** - {***e***} sums to ***T*** - ***e***
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- has ***e*** — ***X*** - {***e***} sums to ***T*** - ***e***
- doesn't have ***e*** — ***X*** - {***e***} = ***X*** sums to ***T***

Algorithm: recurse on ***S*** - {***e***} with ***T*** and ***T*** - ***e*** as the new targets.

Enumerating subsets

```
SubsetSum(S, T):
    if T = 0
        return True
    else if T < 0 or S = ∅
        return False
    else
        pick e ∈ S
        include := SubsetSum(S − {e}, T − e)
        exclude := SubsetSum(S − {e}, T)
        return include or exclude
```

Enumerating subsets

If S is given as an array $S[1, \ldots, n]$, how do we quickly find e and $S - \{e\}$?

Enumerating subsets

If S is given as an array $S[1, \ldots, n]$, how do we quickly find e and $S - \{e\}$?

- $e = S[n]$
- $S - \{e\} = S[1, \ldots, n-1]$

Enumerating subsets

```
SubsetSum(S[1,...,n], T):
    if T = 0
        return True
    else if T < 0 or n = 0
        return False
    else
        e := S[n]
        include := SubsetSum(S[1,...,n-1], T-e)
        exclude := SubsetSum(S[1,...,n-1], T)
        return include or exclude
```

Enumerating subsets

```
SubsetSum(S[1,...,n], i, t):
    if t = 0
        return True
    else if t < 0 or i = 0
        return False
    else
        e := S[i]
        include := SubsetSum(S, i − 1, t − e)
        exclude := SubsetSum(S, i − 1, t)
        return include or exclude
```

Enumerating subsets

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        e := S[i]
        include := SubsetSum(S, i − 1, t − e)
        exclude := SubsetSum(S, i − 1, t)
        return include or exclude
```

Initial call: $\textsc{SubsetSum}(S, n, T)$

Enumerating subsets

$\textsc{SubsetSum}(\boxed{S[1,\ldots,n], i,} t)$:

if $t = 0$

return $\textsc{True}$

else if $t < 0$ or $\boxed{i = 0}$

return $\textsc{False}$

else

$e := \boxed{S[i]}$

$\mathit{include} := \textsc{SubsetSum}(\boxed{S, i-1,} t - e)$

$\mathit{exclude} := \textsc{SubsetSum}(\boxed{S, i-1,} t)$

return *include* or *exclude*

Initial call: $\textsc{SubsetSum}(S, n, T)$

This solution ***indexes*** the subproblems. (We'll see why later)

Enumerating subsets

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        return False
    else
        e := S[i]
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        exclude := SubsetSum(S, i − 1, t)
        return include or exclude
```

$$T(n) = 2\,T(n-1) + O(1) = \Theta(2^n)$$

Backtracking guidelines

1. Identify recursive structure and subproblems (recurrence)
2. Solve the base cases
3. Try all first choices ("take a nibble"), and let recursion take care of the rest.

```
SubsetSum(S[1,...,n], i, t):
    if t = 0
        return True
    else if t < 0 or i = 0
        return False
    else
        e := S[i]
        include := SubsetSum(S, i − 1, t − e)
        exclude := SubsetSum(S, i − 1, t)
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```

Variations

This code only checks if there exists a solution. How can we modify it to output:

- Output the actual solution?
- The number of solutions?
- The “best” solution?

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- Output the actual solution?
- **The number of solutions?**
- The “best” solution?

Pure functions are easy to modify!

Counting the number of solutions

```
SubsetSum(S[1,...,n], i, t):
    if t = 0
        return TRUE
    else if t < 0 or i = 0
        return FALSE
    else
        e := S[i]
        include := SubsetSum(S, i − 1, t − e)
        exclude := SubsetSum(S, i − 1, t)
        return include or exclude
```

Counting the number of solutions

$\textsc{CountSubsets}(S[1,\ldots,n], i, t)$:

if $t = 0$

return $\boxed{1}$

else if $t < 0$ or $i = 0$

return $\boxed{0}$

else

$e := S[i]$

$\mathit{include} := \textsc{CountSubsets}(S, i-1, t-e)$

$\mathit{exclude} := \textsc{CountSubsets}(S, i-1, t)$

return $\boxed{\mathit{include} + \mathit{exclude}}$

Wood cutting

Given a beam of wood ***n*** units long and selling prices ***p***[***i***] for a piece ***i*** units long, what's the most money you can make by chopping up the cord?

Wood cutting

length	1	2	3	4	5	6	7	8	9	10
price	1	5	8	9	10	17	17	20	24	30

Wood cutting

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$10

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$\$10$

$\$5 + \$8 = \$13$

Wood cutting

Sequence of choices: cuts in the piece of wood

Wood cutting

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Base case: length 0, no money

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First choice/nibble: cut out the leftmost piece. How should we cut it?

Wood cutting

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First choice/nibble: cut out the leftmost piece. How should we cut it?

Try each possibility and see which one's best.

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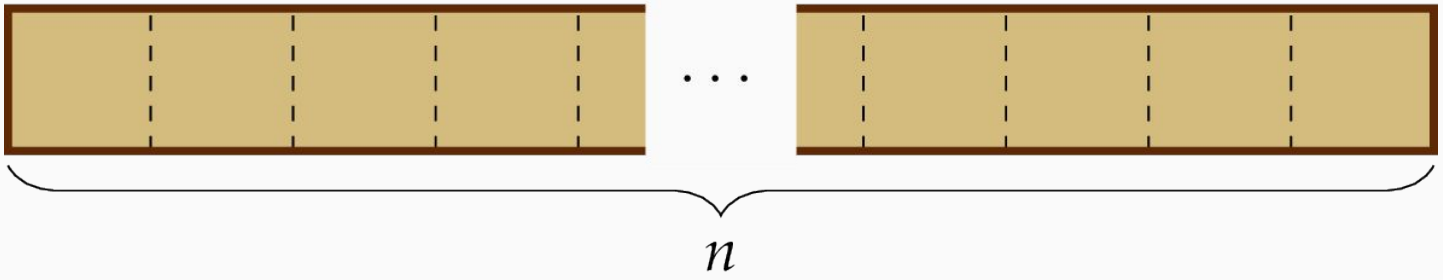
length	1	2	3	4	5	6	7	8	9	10
price	1	5	8	9	10	17	17	20	24	30



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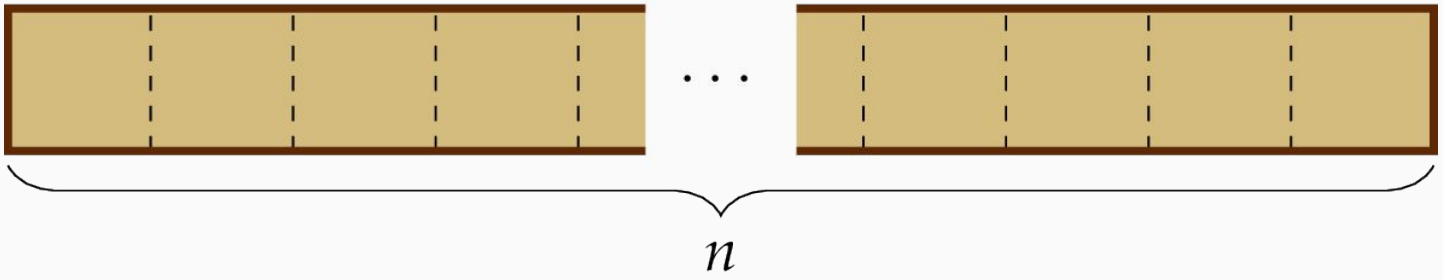
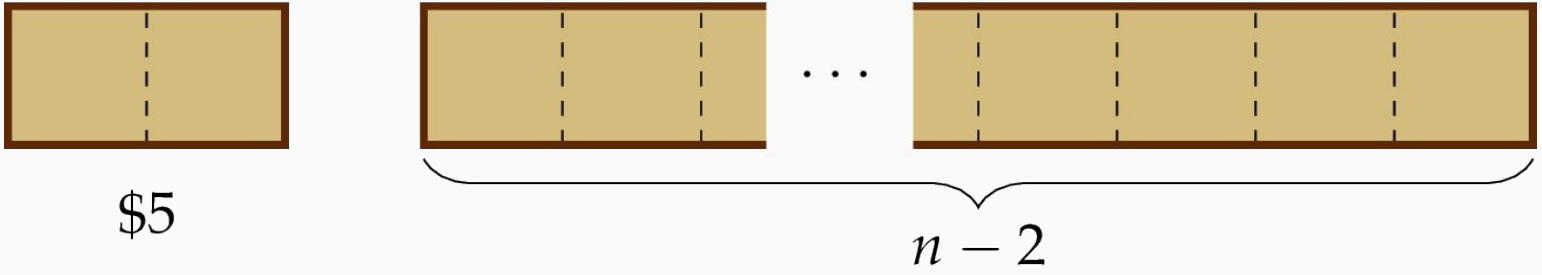
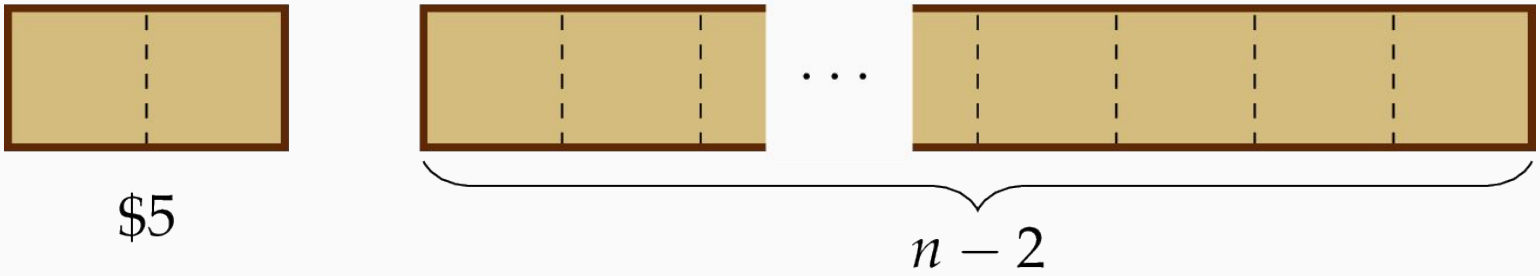


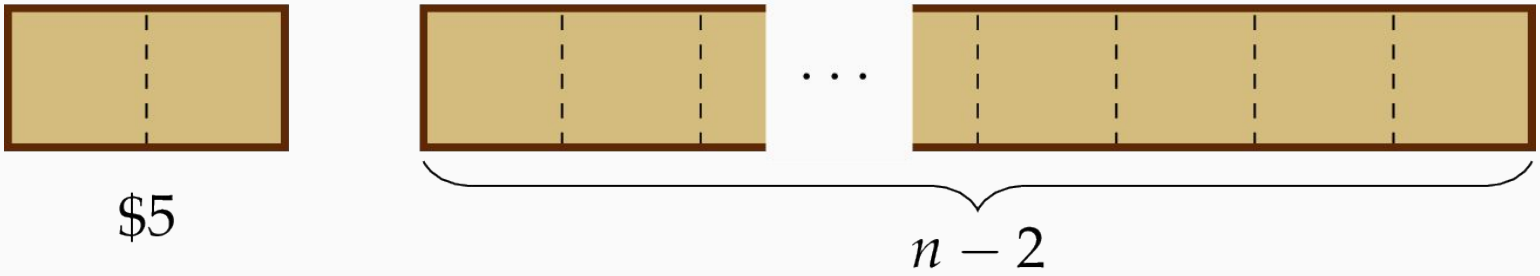


Wood cutting



How do we compute the best profit for what's left over?

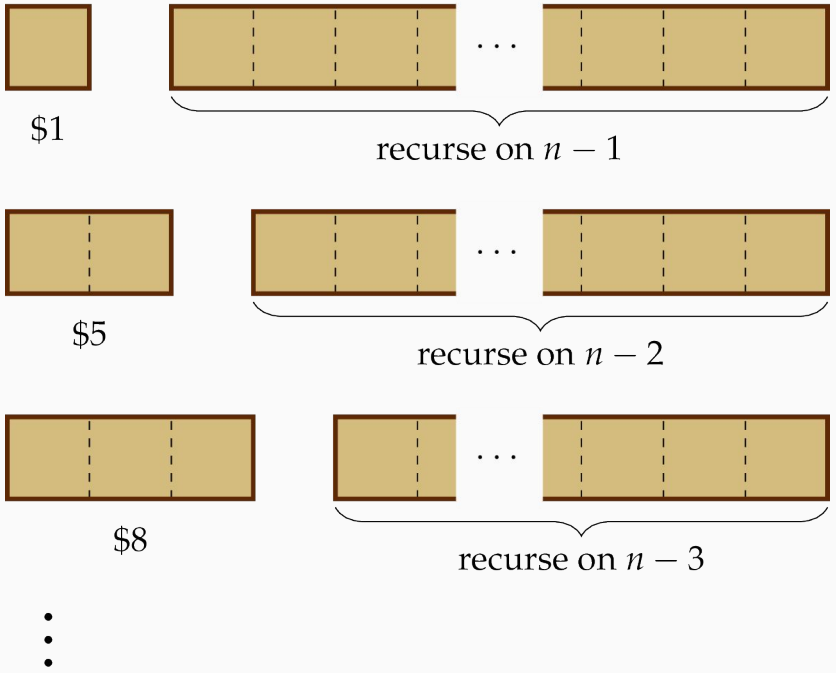
Wood cutting



How do we compute the best profit for what's left over? **Recursion!**

Finding the best solution

Try out all possible “first pieces” and pick the one that maximizes profit.



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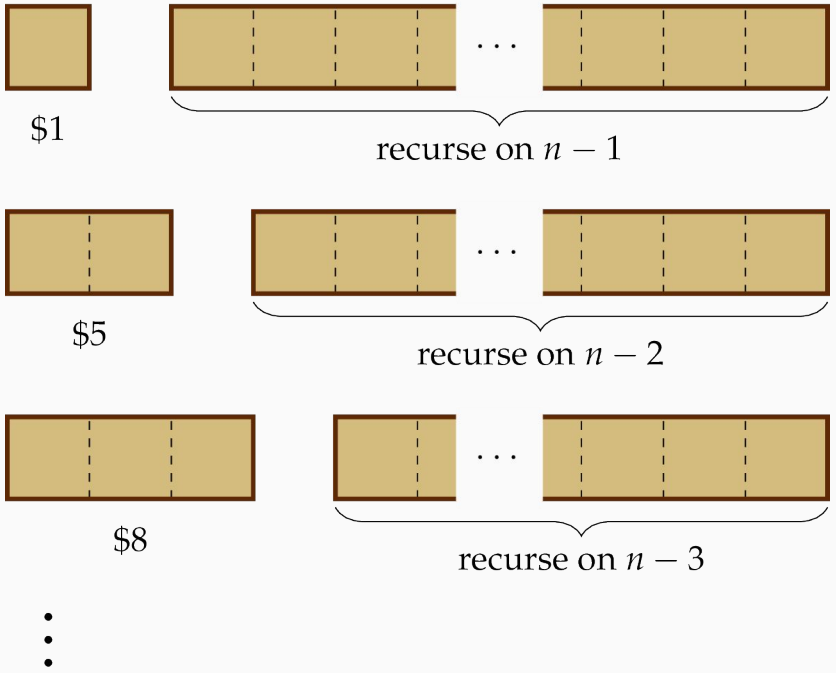

$$P(m) = \begin{cases} 0 & \text{if } m = 0 \\ \max\left(p[1] + P(m-1), p[2] + P(m-2), \ldots, p[m] + P(0)\right) & \text{otherwise} \end{cases}$$

Wood cutting

```
WoodCut(p[1,...,N], m):
    best := 0
    for i = 1,...,m
        profit := p[i] + WoodCut(p, m − i)
        if profit > best
            best := profit
    return best
```

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Wood cutting

$$
\begin{aligned}
&\textsc{WoodCut}(p[1,\ldots,N], m):\\
&\qquad \mathit{best} := 0\\
&\qquad \textbf{for } i = 1,\ldots,m\\
&\qquad\qquad \mathit{profit} := p[i] + \textsc{WoodCut}(p, m-i)\\
&\qquad\qquad \textbf{if } \mathit{profit} > \mathit{best}\\
&\qquad\qquad\qquad \mathit{best} := \mathit{profit}\\
&\qquad \textbf{return } \mathit{best}
\end{aligned}
$$

The base case should be ***m*** = 0, but where is it?

Text segmentation

Given a string of characters, can we insert spaces so that we get a sequence of words?

```
THISISTHEREASON
```

Text segmentation

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```
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Text segmentation

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```
THISISTHEREASON → THIS IS THE REASON

THISISTHEREASON → THIS IS THERE A SON
```

Text segmentation

Possible that some words don’t appear in any valid partition.

TH|**IS**|ISTHEREASON

Text segmentation

Possible that some words don’t appear in any valid partition.

TH | **IS** | ISTHEREASON

No greedy approaches will always work.

Text segmentation

Possible that some words don't appear in any valid partition.

`TH|IS|ISTHEREASON`

No greedy approaches will always work. **Let's solve using backtracking**.

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Sequence of choices: places to add spaces

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Base case: length 0, splittable into 0 words

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First choice/nibble: pick out the first word. What is that first word?

Text segmentation

Sequence of choices: places to add spaces

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First choice/nibble: pick out the first word. What is that first word?

> **Try every possible prefix and see which one leads to a splitting.**

Text segmentation

First word, recurse on rest:

```
T HISISTHEREASON

TH ISISTHEREASON

THI SISTHEREASON

…
```

Text segmentation

```
SPLITTABLE(S[1,...,n]):
    if n = 0
        return TRUE
    for i = 1,...,n
        if S[1,...,i] is a word
            if SPLITTABLE(S[i+1,...,n])
                return TRUE
    return FALSE
```

Text segmentation

BOTHEARTHANDSATURNSPIN

- ~~B OTHEARTHANDSATURNSPIN~~
- ~~BO THEARTHANDSATURNSPIN~~
- BOT HEARTHANDSATURNSPIN
- BOTH EARTHANDSATURNSPIN
- ~~BOTHE ARTHANDSATURNSPIN~~

Text segmentation

What’s the worst-case scenario?

Text segmentation

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- Dictionary: a, aa
- String: aaaaaaaaaaaaaaaa...aaaaaaaaab

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$$T(n) = T(n-1) + T(n-2) + O(1) = \Theta(\phi^n)$$

$$\phi = \frac{1+\sqrt{5}}{2} \approx 1.618\ldots$$